

Activity, part A

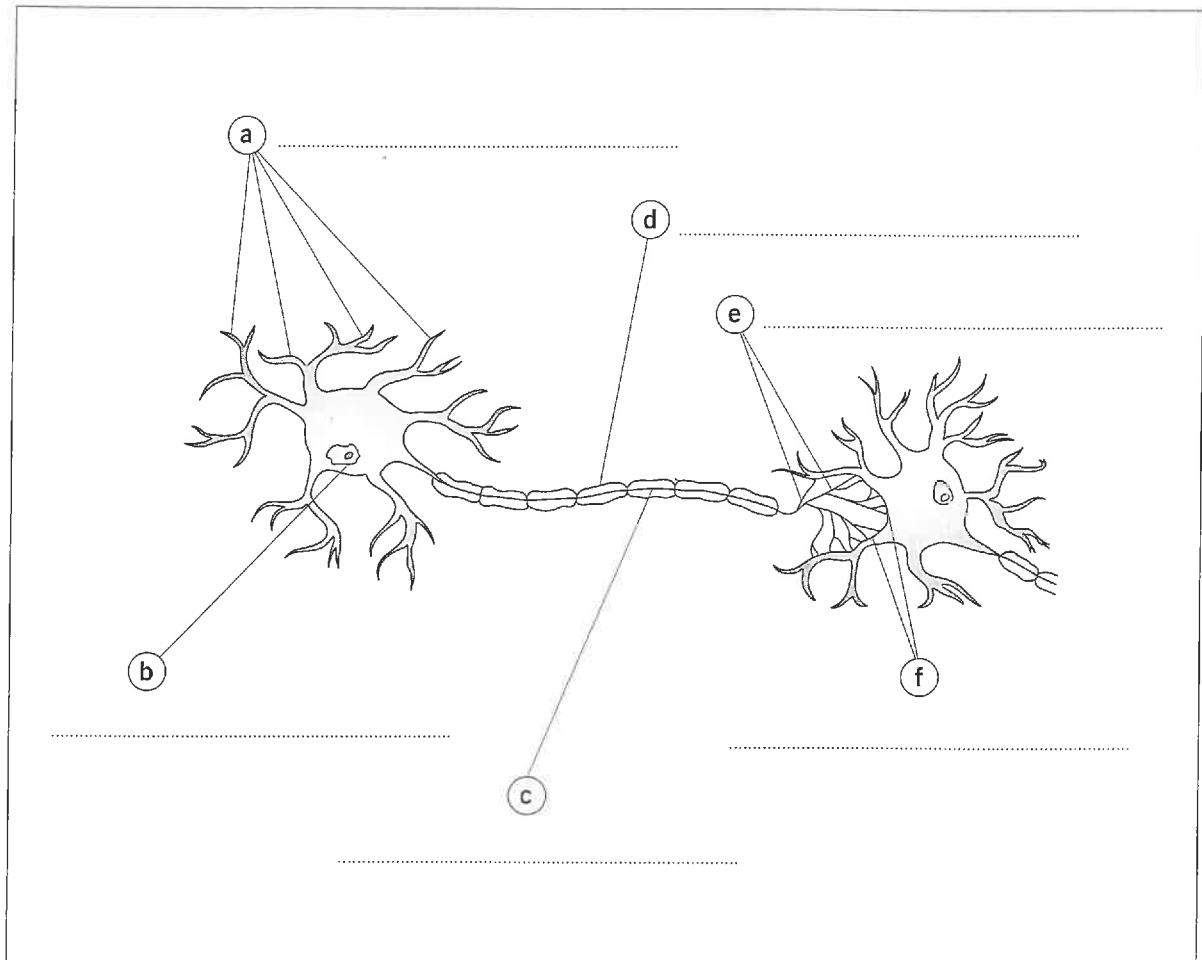


Figure 4.3 A basic neuron

Instructions, part B

- 1 Read the following list of bullet points.
- 2 Write each bullet point under the correct heading in Table 4.2 on the following page. (Hint: The bullets have been placed for you.)
 - transmits information to the soma
 - a single, long, thin fibre that carries information away from the soma
 - the cell body
 - chemicals found in the synapse between neurons
 - branching fibres at the end of the axon that send information across the synapse
 - the electrically charged (50 millivolts) nerve impulse
 - moves down the axon, away from the soma
 - the white, fatty substance that protects the axon and helps transmit the neural impulse along the axon
 - the minute space between one cell's axon terminal and another cell's dendrite
 - sends the neural impulse across the synapse to a dendrite
 - receives incoming information delivered by its dendrites
 - when damaged, loss of sensation or lack of motor control can result
 - filled with neurotransmitters
 - branches (neuron fibres) extending from the soma that receive information from other cells
 - their release can excite or inhibit the activity of a neighbouring neuron

Activity, part B

Table 4.2 Neuron components and neural transmission

axon	• •
dendrites	• • •
soma	• •
synapse	• •
neurotransmitter	• • •
myelin	• •
axon terminals	•
action potential	• •

Activity, part A

Table 4.3 Titles of brain areas and associated structures

Spinal cord	Forebrain	Parietal	Hindbrain
Temporal	Occipital	Midbrain	Frontal

Scissors icon

..... (brain stem)

- lies between the forebrain and the hindbrain
- contributes to reflex responses
- involved in sleep and arousal by stimulating other brain regions

..... (brain stem)

- comprised of the pons, medulla and cerebellum – structures that control basic survival functions (e.g. heart and breathing rate) and reflex actions (e.g. vomiting and coughing)
- coordinates voluntary muscle movement and sequential movement
- involved in sleep and arousal

..... lobe

- the largest and most highly developed brain region
- has two hemispheres
- each hemisphere has four lobes
- contains major structures responsible for our most complex behaviours

..... lobe

- receives and processes visual information

..... lobe

- receives and processes auditory information (sound)
- involved in memory

..... lobe

- transmits sensory information to the brain from the body
- transmits motor messages from the brain to the body
- controls simple spinal reflex actions

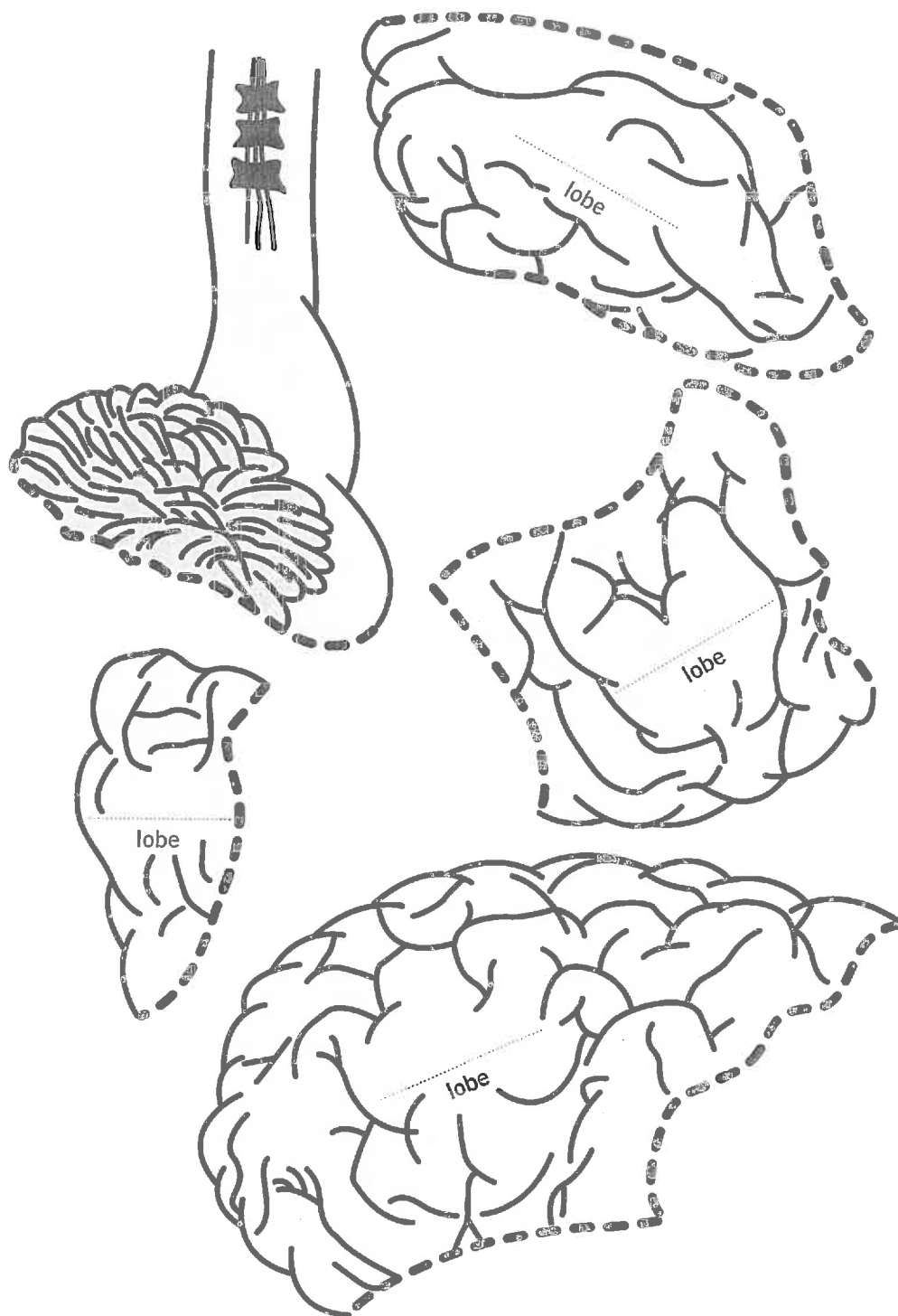
..... lobe

- largest lobe
- centre of consciousness
- responsible for thinking, planning and analysing
- involved in emotions, personality and memory
- controls motor movement

..... lobe

- receives and processes bodily sensations (e.g. touch and temperature)

Figure 4.5 Information about brain areas



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